

From Bean to Bar: A Multidimensional Analysis of Cacao Supply Chains, Commodity Price Volatility, and the Ethical Economics of the Global Chocolate Industry"

Introduction

Chocolate is one of the most favorite delicacies of people around the world. No matter if it's dark chocolate or milk chocolate or even white chocolate, everyone enjoys a good piece of chocolate. It can come in the form of bars or surrounding fruits and pretzels, even in little dollops called truffles. As a liquid it is enjoyed as a variety of milk and even a shake or flavored drink. It is recorded that people in America eat 10-12 pounds of chocolate a year and that isn't even close to the 24 pounds that are consumed in Switzerland. Chocolate is actually known as *Theobroma cacao* and comes from cacao beans so technically chocolate is not the crop, the beans evolve into the sweet and dessert everyone loves (Bilton). Over the next 5 chapters we are going to go over the history of chocolate along with how it's made and maintained. We will start from the beginning and focus on the process of growing the crop and the cost of keeping its production thriving.

History and Origin

As one of the world's favorite foods, chocolate has a rich history. Before chocolate was known worldwide, in South America, the Aztecs, Olmecs, and Mayans were very familiar with cacao beans. At the time the beans were used as currency and even included as offerings to their gods. However when it was prepared as a food or drink it was not as delicious as it is this day. The

Mayans actually ground the beans with maize and mixed it with water, often adding vanilla and an ingredient called ear flower (*Cymbopetalum penduliflorum*) (Bilton). The chocolate drinks we know today are sweet and rich with flavor but this drink that the Mayans served was savory and often has spices similar to black pepper. The Aztecs had their own version which meant boiling the mixture and adding chilli powder which is most similar to the hot chocolate we have today. Europeans in 1575 who visited the land were not fond of the cacao beans as they believed it was a drink for pigs. However when they couldn't drink wine they determined it was a refreshing drink. Chocolate was soon spread across the globe as the Spanish had already created recipes to make the cacao beans more suitable for one's taste. Unlike the ancient civilizations, they added sugar and formed the cocoa into cakes (Bilton). Once these cakes were formed they would add water to make the chocolate drinkable. As the beans were not known, some in England believed they were sheep droppings and torched cargo of beans they had seized in 1579. Only a little more than 75 years later did the English know the value of the beans when they colonized Jamaica and their plantations of cacao beans. Chocolate was now an expensive delicacy due to importing fees so only the rich would afford the beans. The English were the first to incorporate milk to their recipe of chocolate "drink". They added sugar and sometimes a hybrid of milk and water to some egg yolks and a little amount of alcohol. As they usually drank their tea in plates, they added their chocolate drink to that tradition and drank their new recipe from plates as well. Chocolate became even more popular after it was said to have health benefits (Bilton). Some said that it prolonged life and provided the same amount of nutrients as a pound of beef, but easy to digest. Coffee houses in England began to serve chocolate and then shops dedicated to chocolate were established, Chocolate Houses. Although chocolate drinks are quite famous, the creation of the chocolate bar was revolutionary. In the early 19th century, a dutch chemist named Coenraad

Johannes Van Houten shared a process of grounding the beans into a powder which made it much easier to digest since mixing the water was more smooth (Bilton). Soon in the 1850s, a Quaker chocolate company began to mix sugar with powder and cocoa butter to make a paste that could be shaped into bars. This invention catapulted the chocolate business into a successful industry that led to companies like Fry and Sons to supply chocolate to the Royal Navy and eventually caused the establishment of Cadbury who were chosen to supply chocolate to Queen Victoria (Bilton). To this day chocolate sales do not slow as it continues to be a major food in every home.

Where Its Grown

As of 2019 the top 10 countries that grow cacao actually grow more than 75% of the entire world's cacao and 50 additional countries take part in the production too (What Countries Grow Cacao?, 2024). There is a difference: Chocolate is manufactured the majority of the time in Europe and America. But cacao is grown in Africa and South America. Cacao needs to be grown in humid environments so that is why it is not grown in America or Europe. Although we get really good quality chocolate from Switzerland and Belgium, the cacao that is used in making the chocolate in those countries are coming from Africa and Central America. Many factories until the 3rd quarter of the 20th century manufactured their own cocoa and chocolate (Beckett, 2018). However the manufacturing was then done by several large operations and then smaller companies started buying their cocoa from other suppliers (Beckett, 2018). Now let's cover the growing process! Cocoa needs to be grown in a specific way since it is a fragile plant. The beans need sufficient rain and shade but they must grow in a warm climate. Therefore the countries it's

grown in have a similarity: They are within 20 degrees of the equator on both sides so they can get the most yield (Cocoa Growing, 2024). Now let's go over the growing process! Just like other processes of planting, farmers must maintain the soil and care for the trees. Unlike other plants, cocoa trees take 4-5 years to yield the pods but they continue to produce them for multiple decades after (Cocoa Growing, 2024). A pod usually has 30-40 beans and a tree habitates 30 (Cocoa Growing, 2024). After the beans are recovered, they are covered and dried so they can ferment which boosts the flavor of the cocoa (Cocoa Growing, 2024). After being stored, they are shipped to manufacturers where they are turned into the cocoa we use today!

How Much We Pay For Chocolate

As much as we love chocolate it is becoming more expensive. Back then it used to be almost 50 cents for a bar but now it is more than double. It all starts on the cocoa trees. Since cocoa trees follow a specific set of rules for them to grow efficiently, there is a limited amount of area for them to grow. This leads to forests being destroyed so more plantations can be built for the trees. And since weather changes and the land becomes very limited, cocoa will be harder to grow and therefore more expensive to buy. Cocoa farmers are losing plantations due to these factors so the next time you buy chocolate, pay a little extra for ethically sourced cocoa since it can be agreed that it's an expensive delicacy (Khan, 2021). A century ago, chocolate was known as a cheap food but now you can find chocolate as expensive as \$300 a bar (*Why the Most Expensive Chocolate in the World?*, n.d.). Some chocolate connoisseurs use rare cocoa and therefore charge a very high price to those who wish to purchase it. While some chocolate bars are a mere \$ $\frac{2}{3}$ a bar, many people indulge in more expensive chocolate in which the cocoa is grown in very

distinct parts of the world and has a stronger flavor. Cheaper chocolate bars utilize cheaper cocoa that is commonly grown and easier for people to buy. But now some are saying to compare chocolate to wine as the process to make it is as specific and can be very high in costs (*Why the Most Expensive Chocolate in the World?*, n.d.) (Khan, 2021). Obviously not everyone has a large budget for chocolate so stick to the chocolate bars you love. No matter what the Guinness world record is for a chocolate bar's cost, even a \$1 chocolate wafer is enjoyed around the world.

Diseases, Weeds, and Insects and What Farmers Can Do to Control the Pests (2 questions in 1)

There are always troubles when it comes to growing any plant. All types of pests can harm the growth of cocoa trees and eventually have an effect on the manufacturing of cocoa and chocolate. An important disease that can affect the growth of cocoa trees and the harvest is called *Phytophthora palmivora*, also known as pod rot. The symptoms include the pod translucent spots which then become dark and firm leading the pod to turn completely black (Plant Village). The cause for this disease is called Oomycete which is a fungus in the form of eukaryotic microorganisms. "Mummified pods" are the ones that succumb to this disease and should be removed because it can be contagious. To help treat the disease or be proactive, farmers should spray coats of copper that have fungicides which can control the rot. And in general the trees should be spread out to not risk spreading the disease and help with the circulation of the air (Plant Village). A virus can also affect the growth of the cocoa trees. Swollen shoot virus can cause leaves to be swollen and can smear the color which can eventually lead to the tree dying

(Plant Village). Just like the previous disease the proximity of the trees is significant. Trees should be at a distance and the plantations should be isolated. Insects can also damage cocoa trees. Cocoa mealybugs can attract ants on the trees and the colony can lead to fungus (Plant Village). Even though there are no clear signs of significant damage, the mealybugs presence could possibly transmit a viral disease like the swollen shoot virus (Plant Village). Just like with other insects, chemicals can help manage the mealybug infestations. Although land is limited some suggest growing other plants too which can naturally help separate the cocoa trees and the circulation of the air. Since wind is a factor that can destroy the trees, they must be grown in a calm environment that is well shaded but also humid as emphasized before. It is important to isolate the trees that are not affected by disease and to dispose of rotting or mummified pods. Diseases can destroy entire farms which is a big waste for the harvesters, keep in mind that it takes multiple years to yield the pods. Lots of hard work goes into growing the trees so that's why the chocolate can be expensive and even difficult to manufacture if the growing conditions are not suitable for the cacao. Jumping forward to after the chocolate is made and you have bought it from a store, you can keep it in a cool dry place so it does not melt. And like other foods, do not leave it out in the open air otherwise it can mold. Some forms of chocolate may be served at different temperatures ex: hot chocolate and mole. Overall many steps can be taken to ensure the best quality of cocoa is grown.

Market and Transport

Cocoa and chocolate are sold around the world at various monetary values. After cocoa is let out to dry under banana leaves, they ferment and are collected. They are cleaned and defected beans

are sorted out. Intermediaries who represent chocolate manufacturers buy the beans directly from the farmers (World Cocoa Foundation, 2024). Afterwards the bags are moved to an exporting business that goes through all the cocoa to determine the quality. Sometimes the beans need to be dried more so they will do that there before they're sent to a warehouse near a port. This is not always the sequence of moving the beans as sometimes the beans are directly bought by the exporter and/or shipped (World Cocoa Foundation, 2024). After the warehouse step they are loaded into cargo vessels in something called a jute bag. Shipping the beans this way is cheaper and in bulk (World Cocoa Foundation, 2024). After the beans are processed into powder and made into various chocolate delicacies, it is time to market the food. The chocolate is then sold in stores and online to people around the world at different values. Some brands have their own stores which only serve chocolate so it is easier to reach their target market. There are different categories when it comes to chocolate. They can be sold as milk, dark, white or they can be sold in terms of season or the quality of the cocoa (World Cocoa Foundation, 2024). Once they hit the packaging stage, chocolate must be sold in containers that make customers want to buy it. They will often have a lot of color and the fonts of the chocolate will be fancy and sometimes in cursive. They will utilize gold and include multiple flavors in one container to offer variety. They will advertise via commercials and billboards, often including celebrities in the marketing strategy. Brands will sell their chocolate in different forms. For example truffles and bars are very popular. They will collaborate with other brands and certainly use the different types of chocolate (white, milk, dark) to let the consumer have more choices. Chocolates from Europe get the most attention as it is reputable and of good quality, therefore many brands emphasize the part of Europe the chocolate is manufactured in. Needless to say chocolate does not need to be this marketed but considering the brands are making lots of money, they can definitely help boost

their advertising and expand their market. My favorite chocolate is Lindt which comes from Switzerland. However it is grown by the hardworking farmers in Africa and Central and South America. Lindt is known for their truffles which have various flavors and every so often they will drop a new type of chocolate. This can help boost their sales and add a new chocolate of theirs to the market. Chocolate is used in food too! Mole is a chocolate sauce from the state of Puebla. This Mexican chocolate sauce is cooked with chillies to give it a spicy flavor. On the less spicy side, hot chocolate is a common drink during cold weather. Some add marshmallows to the drink and unlike the old times, most people use milk as the base instead of water. In France they serve you chocolate with whipped cream and not all establishments that serve hot chocolate add marshmallows. For those who are vegan, chocolate is available to them as well and doesn't utilize any dairy products. For example some dark chocolate is naturally vegan since it doesn't contain milk. Chocolate can be enjoyed by all and will continue to be one of the most loved foods ever!

Conclusion

Over these chapters I am sure you know now the amount of effort it takes to make a single chocolate bar. The number of years it takes for one tree to grow until it can yield beans, the amount of obstacles that need to be jumped over so the quality does not decrease, and the chain of events for the chocolate to be given in our hands takes a large amount of time. The next time you go buy a chocolate bar keep in mind the adventure the cacao went on so it could be delivered to you!



Source: Reuters

Fig. 1 A field worker in Ghana points out a leaf affected by the swollen shoot virus. You can see the holes in the leaves (Angel, 2024).



Source: Ecuador & Cocoa

Fig. 2 Cacao beans are collected and covered with a blanket or banana leaves to start the fermentation so they can be dried and sold to manufacturers (*Ecuador & Cocoa*, n.d.).

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